Fine-tuning collapses differentiation gradients in single-cell foundation models in a domain-specific manner

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Abstract

Supervised fine-tuning of single-cell foundation models is routine practice, yet whether it preserves the continuous differentiation geometry in pretrained representations has not been quantified. Using the human corneal epithelium, a four-stage differentiation continuum (limbal stem cells, transit-amplifying cells, basal, and differentiated) with known ordinal structure, we show that classification fine-tuning of Geneformer-V2-104M disrupts the differentiation gradient domain-specifically: catastrophic collapse on same-corpus data (Spearman ρ 0.806 $\rightarrow -0.008$) with preserved layer topology, mild collapse on truly out-of-corpus data (0.707 $\rightarrow$ 0.641), and improvement on cross-tissue data (0.24 $\rightarrow$ 0.46). An ordinal head (CORN) is the only recipe preserving ρ (0.833) at matched accuracy, and only within-corpus; five alternative objectives all failed. Primary claims survive BH-FDR control (26 comparisons) and label/feature-shuffle nulls. We release a contamination-aware benchmark pairing same-corpus and out-of-corpus external tests.

Keywords: single-cell foundation models, classification fine-tuning, differentiation gradient, ordinal regression, contamination-aware benchmark, corneal epithelium

1. Introduction

The corneal epithelium is maintained by a tightly regulated differentiation continuum from quiescent limbal stem cells (LSCs) at the limbal palisades of Vogt, through transit-amplifying cells (TACs), to committed basal epithelial cells, and ultimately to terminally differentiated squamous cells forming the protective ocular surface barrier [1]. This ordered progression is functionally required: disruption at any stage (LSC deficiency, aberrant amplification, or premature terminal differentiation) compromises corneal transparency and may lead to irreversible vision loss [2].

Single-cell RNA sequencing (scRNA-seq) has resolved this continuum at single-cell resolution, with marker genes for LSCs (e.g., *TCF4*, *POSTN*, *ABCG2*, and *TSPAN7* [3]; *TP63* and *KRT15* [4, 5]), TACs (e.g., *UBE2C*, *CDC20*, and *CENPF* [6]; *MKI67* and *BIRC5* [3]), and differentiated populations (e.g., *KRT12* and *KRT3* [4, 7, 8]) identified across independent studies and aggregated in community atlases such as the human ocular surface atlas in

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CELLxGENE Discover [9]. Yet these efforts produce study-specific embeddings that are not directly comparable across datasets, and no unified reference model maps individual cells to a shared, ordered differentiation-state space. The closest approach, a corneal cell-state reference model with support-vector-machine classification (cPredictor) [10], provides a supervised classifier rather than an evaluation of foundation-model representations or their geometric structure.

Foundation models pretrained on massive single-cell corpora offer a potential solution. Geneformer, a 12-layer Transformer pretrained on approximately 104 million cells from 2,903 datasets [11, 12], and related architectures such as scGPT [13] and scBERT [14], learn contextual gene–gene representations that transfer across tissues. Systematic evaluations have tempered expectations: a unified benchmark of thirteen single-cell foundation models found classical PCA remains competitive in several settings, with pretrained embeddings advantageous mainly in low-label and transfer-learning scenarios [15]; biology-driven re-analyses questioned the pre-train-then-fine-tune paradigm [16]; and large-scale evaluations found that, for cell-type annotation, frozen or zero-shot usage generally underperforms fine-tuning, and freezing pretrained layers was not recommended [17]. Conversely, foundation models designed for zero-shot use can organize developmental lineages and differentiation hierarchies without task-specific training [18], indicating that pretrained representations can encode continuous differentiation structure. Benchmark-driven evaluation has become an established genre in this journal, spanning single-cell foundation models [19, 20] and clinical question answering [21], and representation-quality assessment is emerging as a distinct concern in embedding evaluation [22]. Whether general-purpose representations adequately capture the ordered structure of a specific differentiation hierarchy, such as the corneal continuum, remains open; conversely, task-specific fine-tuning may improve discriminative accuracy while inadvertently collapsing the continuous geometric relationships among adjacent differentiation states that are biologically essential.

The fine-tuning literature has focused on whether adaptation improves classification accuracy and whether it causes forgetting of learned cell-type knowledge; full fine-tuning induces catastrophic forgetting at the label level, motivating parameter-efficient remedies [23]. With respect to geometry, a zero-shot benchmark reported geometric collapse of foundation-model embeddings during cellular dynamics reconstruction, with HVG/PCA outperforming them for trajectory recovery [24], and EMP-FM showed that naive fine-tuning often fails to resolve intermediate cell states under a dual classification–pseudotime objective [25]. Layer-resolved dissections of single-cell foundation models have begun to probe where biological information resides within the network [26, 27], and fine-tuned representations are already deployed in production annotation protocols without quantifying whether adaptation preserves ordinal structure [28]; a related cautionary literature documents cases in which well-performing models fail in settings that matter [29, 30]. What remains absent is a quantified measurement, under controlled fine-tuning conditions, of whether supervised adaptation preserves or destroys the continuous differentiation-state gradient that orders a hierarchy, which we term the *continuity problem*. The corneal epithelium is an ideal test system: its four ordered stages have well-established markers, so the expected ordinal structure is known *a priori* and gradient preservation can be measured directly in embedding space. We evaluate a family of candidate gradient-preserving fine-tuning recipes under an identical locked protocol, including a continuity-regularized objective (CAFT), a cumulative ordinal head (CORN), an

axis-anchored objective (AA-FT), a soft-Spearman rank surrogate, and a flow-consistency objective; whether any of them preserves ordered geometry at matched classification accuracy is a hypothesis to be tested on externally held-out, source-disjoint data, not an assumption.

Here we quantify the effect of classification fine-tuning on differentiation-state geometry in the normal human corneal epithelium. Using a donor-disjoint fair-split protocol on a 10-study corneal and ocular surface atlas (99,606 training cells from 49 donors; 20,394 test cells from 25 donors; zero donor overlap), we compare two reference arms (HVG/PCA with a linear classifier; frozen Geneformer-V2-104M with a linear probe) with adapted configurations sharing identical preprocessing, data splits, classification head, and training budget: naive cross-entropy fine-tuning, a continuity-regularized objective (CAFT), an ordinal head (CORN, with and without the continuity term), and three further gradient-preserving objectives (AA-FT, soft-Spearman, flow), evaluated on a same-corpus donor-disjoint holdout (GSE153515; 17,587 cells, 2 donors), a truly out-of-corpus external test (GSE299152; 21,374 epithelial cells, 3 donors), and a cross-tissue epidermal test set. Our contributions are fourfold: (1) a quantified, three-domain measurement of fine-tuning-induced differentiation-gradient collapse: catastrophic on same-corpus data (the *topology-preserving ordinal-axis collapse*, TP-OAC, layer-resolved across all thirteen hidden states), mild on truly out-of-corpus data, and reversed on cross-tissue data; (2) an ordinal head reparameterization (CORN) as the only recipe preserving the gradient at matched classification accuracy, and only within the pretraining corpus; (3) a systematic honest-negative evaluation of five alternative gradient-preserving objectives, in the spirit of recent critical-evaluation studies in this journal [30, 31]; and (4) a contamination-aware scFM benchmark for the normal human corneal epithelium, pairing a same-corpus donor-disjoint holdout with a truly out-of-corpus external test under a locked multi-seed protocol with BH-FDR control and permutation nulls. These measurements define a collapse mode we term *topology-preserving ordinal-axis collapse* (TP-OAC), in which the ordinal differentiation axis is erased while discrete layer topology survives; it is distinct from the ordinal geometry that ordinal supervision induces [32], from pretrained layer profiles characterized without adaptation [33], and from the trajectory-fidelity trade-offs that trajectory benchmarks quantify across embedding methods [34]. Unlike prior evaluations of embedding quality, which focused on visualization stability [22] or zero-shot geometry [24], our measurements quantify how supervised adaptation itself alters ordinal differentiation geometry under controlled fine-tuning conditions. Our scope is restricted to normal human corneal epithelium; evaluation of disease states, including dry eye disease, is deferred to future work.

2. Results

2.1. Fine-tuning collapses the differentiation gradient while preserving layer topology

Classification fine-tuning collapsed the differentiation gradient of the corneal epithelium while leaving layer topology intact. Two non-adapted references (HVG/PCA with a linear classifier; frozen Geneformer-V2-104M with a linear probe) were compared against end-to-end cross-entropy fine-tuning and a continuity-regularized variant (CAFT), all sharing identical preprocessing, data splits, classification head, and training budget, on a same-corpus, donor-disjoint holdout (GSE153515; Figure 1; three independent seeds per adapted arm; per-seed values in Table S1).

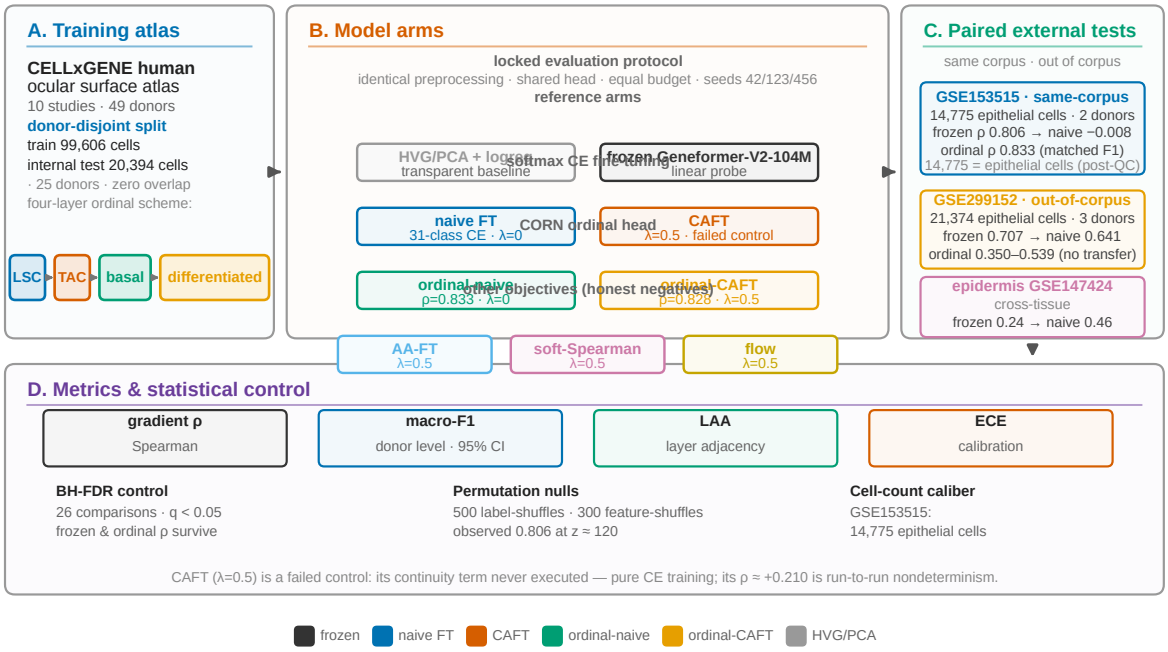


Figure 1: Study pipeline. Donor-disjoint training atlas (CELLxGENE), locked evaluation protocol, softmax-head and ordinal-head fine-tuning arms, and paired same-corpus (GSE153515) and post-cutoff out-of-corpus (GSE299152) external tests.

The two non-adapted reference arms preserved strong differentiation-gradient rank correlation on the same-corpus holdout: Spearman ρ was 0.8206 for HVG/PCA and 0.8064 for frozen Geneformer (Table 2). By contrast, naive supervised fine-tuning collapsed this ordinal signal to near zero: per-seed ρ ranged from -0.042 to $+0.054$ across seeds (three-seed mean -0.0082), with bootstrap 95% confidence intervals spanning zero in every run (Figure 2, Table 2). The sign of ρ was inconsistent across seeds: the residual ordinal correlation after fine-tuning was indistinguishable from noise.

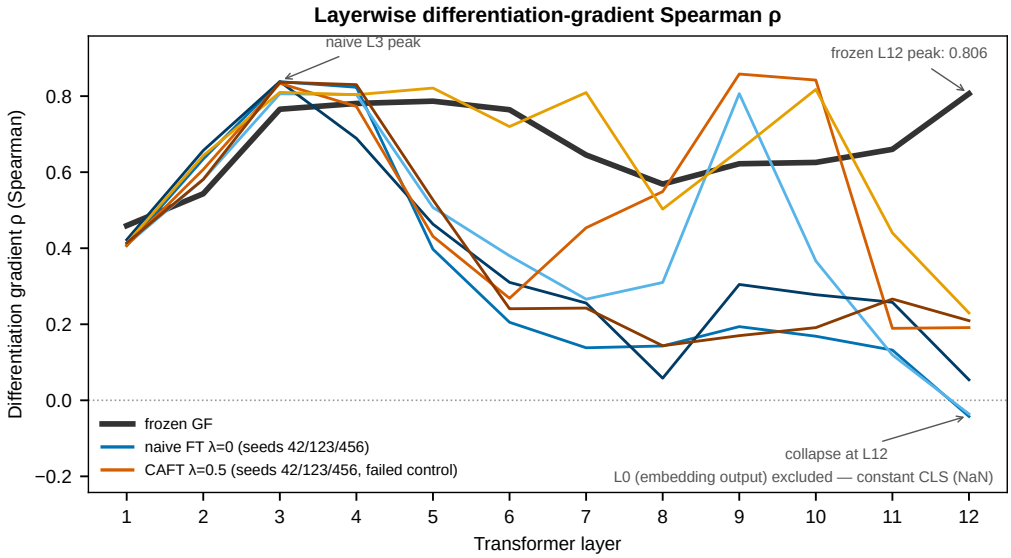


Figure 2: Layer-resolved analysis localizes the collapse to upper transformer layers. Layer-resolved differentiation-gradient Spearman ρ across hidden states L1–L12 for seven Geneformer-based configurations (n = 14,775 external-test epithelial cells; three seeds per adapted arm; L0 excluded: constant [CLS]). Color key: thick dark gray line, frozen Geneformer; blue family, naive fine-tuning (seeds 42/123/456); vermillion family, CAFT (failed control; seeds 42/123/456). The collapse is head-proximal: early layers retain the gradient, while L5–L12 decay. See also Figure S7 and Table S5.

Despite this gradient collapse, discrete layer topology was preserved. Layer adjacency accuracy (LAA) remained high in all four arms: 0.9771 (HVG/PCA), 0.9665 (frozen Geneformer), 0.9768 (naive, three-seed mean), and 0.9773 (CAFT, three-seed mean). Classification fine-tuning therefore disrupted the continuous ordering of the differentiation continuum while leaving local layer-adjacency structure essentially intact: continuous gradient geometry and discrete layer topology are dissociable under task adaptation.

The continuity-regularized arm must be read as a failed control. A post hoc audit of the archived training code established that the continuity term never executed in any run, because of an implementation defect in the training hook, so those checkpoints trained pure cross-entropy, identical in objective to the naive arm (Methods, Provenance and contamination audit). The $+0.210$ separation in final-layer ρ therefore reflects run-to-run nondeterminism rather than a causal effect of the regularization, and classification performance was likewise indistinguishable: macro-F1 was 0.6451 (95% CI 0.6164–0.6738) for the continuity-regularized group versus 0.6492 (95% CI 0.6233–0.6751) for naive fine-tuning ($\Delta = -0.0041$; overlapping intervals). Throughout, we report this arm as an honest negative implementation result

rather than a promoted safeguard. Fine-tuning nonetheless achieved the highest classification accuracy while collapsing the gradient to noise: differentiation geometry and classification performance are separable.

Layer-resolved analysis localized the collapse to upper transformer layers. To determine whether fine-tuning erases the differentiation-gradient signal globally or damages it in a layer-localized manner, we re-measured ρ at all thirteen hidden states (embedding output L0 and layers L1–L12; L0 excluded because the position-0 [CLS] representation is constant across cells) for all seven Geneformer-based configurations on the 14,775 external-test epithelial cells; the layer-resolved pipeline reproduced each model's single-value measurement exactly (Methods, Layer-resolved gradient measurement).

The collapse was concentrated in head-proximal layers, with the gradient intact in early layers (Figure 2). First, early-layer representations preserved the gradient in every arm: L1–L4 ρ lay between 0.41 and 0.84, and the fine-tuned models reached their maximum at L3 (0.81–0.84 across seeds), matching or exceeding the frozen model at the same depth (L3, 0.7656). Second, the collapse observed at the final layer of the naive models was concentrated in the upper half of the network: from L5 (0.3973, seed 42) ρ decayed toward zero or negative values at L12, whereas the frozen model's gradient signal rose to its network-wide maximum of 0.8064 precisely at L12. Third, the continuity-regularized checkpoints ended at L12 ρ of 0.19–0.23, above the naive group's near-zero values but far below the frozen model's 0.8064; as the continuity term never executed (Methods), this difference is not attributable to the regularization objective.

Bootstrap confidence intervals at L12 (n = 1,000, cell-level resampling) quantified the separation: frozen $\rho = 0.8064$ [0.7918, 0.8164]; all naive seeds spanned zero; and the continuity-regularized checkpoints were positive (0.19–0.23) and, in this sample, separated from both groups. Because the continuity term never executed in that arm, the separation is a property of these particular cross-entropy runs, not evidence of a regularization effect. Gradient collapse under classification fine-tuning is therefore a head-proximal overwrite: the signal persists in intermediate layers and is selectively disrupted in the layers nearest the classification head.

To test whether this overwrite reflects the training process rather than an artifact of the trained classification head, we replaced the trained head of the naive seed-42 checkpoint with a freshly initialized, untrained random head: the layerwise differentiation-gradient ρ profile remained bit-identical at all thirteen layers (L12 $\rho = -0.0424$), while the same treatment on the frozen body preserved the frozen profile exactly (L12 $\rho = 0.8064$). The L5–L12 collapse is therefore intrinsic to the fine-tuned body weights, a training-induced rewrite of head-proximal layers, and not a readout artifact of the trained head (Methods, Layer-resolved gradient measurement).

Trajectory inference confirmed the geometric dissociation at the level of inferred cell ordering (Figure 3). Diffusion pseudotime on the same external cells, rooted at the LSC class, correlated strongly with the reference layer-ordering pseudotime for the frozen model ($\rho = 0.850$, 95% CI 0.842–0.858) but weakly for naive fine-tuning ($\rho = 0.323$–0.348), with CAFT intermediate ($\rho = 0.372$–0.428). The loss of ordinal information under naive fine-tuning is thus observable not only in our gradient metric but in a widely used trajectory-reconstruction procedure.

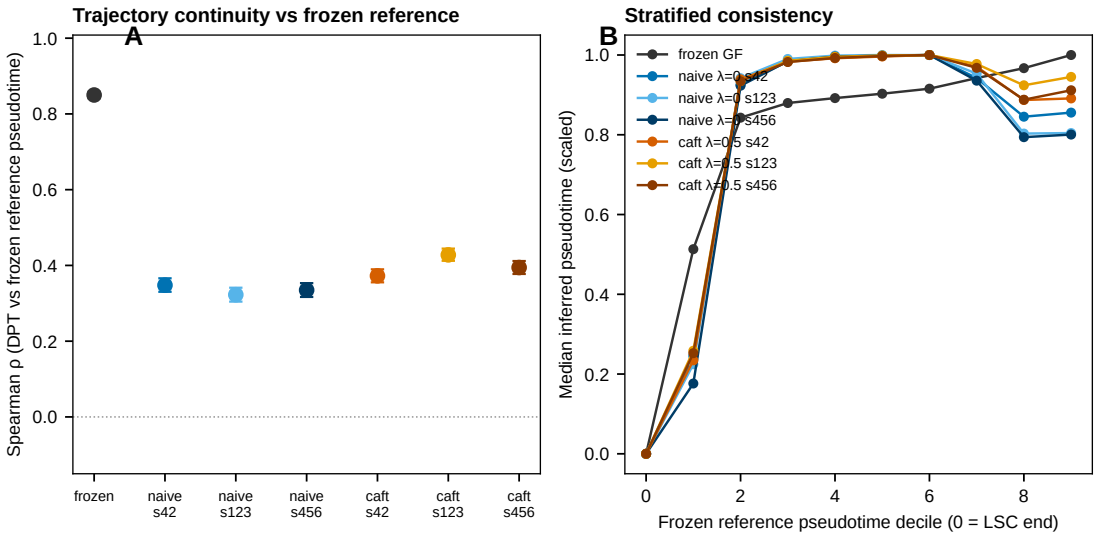


Figure 3: Trajectory inference confirms the geometric dissociation. Diffusion pseudotime on external cells (n = 14,775; three seeds per adapted arm; error ranges are bootstrap 95% CIs, Methods): frozen representations preserve cell ordering relative to the reference pseudotime ($\rho = 0.850$), naive fine-tuning does not ($\rho \approx 0.33$), and CAFT is intermediate ($\rho \approx 0.40$).

2.2. The collapse is domain-specific: a three-domain boundary map

Having measured the collapse on a same-corpus holdout, we asked whether it reflects the evaluation domain rather than fine-tuning itself, and delineated its boundary across three domains of increasing distance from the pre-training corpus. To separate the effect of fine-tuning from the effect of pre-training exposure, we paired two external test sets with opposite corpus status and added a cross-tissue control: GSE153515 (normal limbal epithelium), whose primary publication DOI resolves to a row in the official Genecorpus-104M source table (53,290 cells, organ = eye) and thus lies within the pre-training corpus (least conservative); GSE299152, a single-cell map of the ocular surface glycocalyx [35], deposited after corpus assembly and absent from the table (post-cutoff, most conservative); and human epidermis (GSE147424 [36]) as a cross-tissue control. Construction and annotation of these sets follow the locked protocol in Methods (Figure 4).

On the same-corpus holdout, fine-tuning drove performance while collapsing the gradient. On GSE153515, the HVG/PCA reference arm achieved donor-level macro-F1 of 0.6192 (95% CI 0.5747–0.6637), outperforming frozen Geneformer (0.4421; 95% CI 0.3749–0.5093), the opposite of the internal training-split ranking (0.8834 versus 0.7787). Frozen Geneformer showed poor calibration on external data (ECE 0.3010) and basal-class collapse (recall 0.216), whereas HVG/PCA remained well calibrated (ECE 0.0117) with strong gradient correlation. Both fine-tuned arms (naive 0.6492; CAFT 0.6451) outperformed both baselines. These results suggest that fine-tuning, not pretraining alone, drives cross-dataset generalization (Figure 5); the magnitude of these scores sits in the range expected for donor-disjoint validation, in which external macro-F1 falls well below internal-split numbers across the single-cell foundation-model literature [15, 16]. On this same-corpus holdout, the reference arms preserved the differentiation gradient (frozen $\rho = 0.8064$), whereas naive fine-tuning collapsed it to -0.008

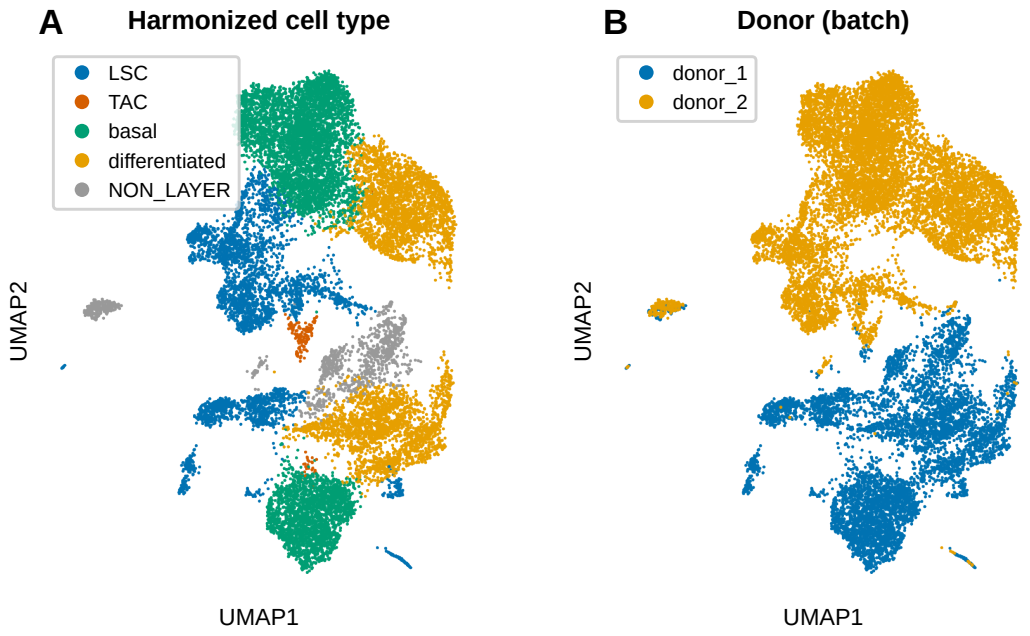


Figure 4: Annotation of the external test sets. External test annotation (GSE153515; n = 15,827 QC-passing cells from two donors): unsupervised integration and clustering, and DEG-based assignment to four ordered differentiation classes (panels A–B). Cluster-level annotation detail (marker dot plot and assignment heatmap) is provided as Figure S9.

with preserved layer topology (TP-OAC, above).

Both adapted arms generalized across the two donors (donor_1 > donor_2 in all six runs; Table 2). Because the TAC class is extremely rare (171/14,775 cells; 1.2%) and received zero true-positive predictions in every adapted run, we performed a sensitivity analysis excluding TAC (Supplementary Figure S10). Three-class macro-F1 rose by +0.21–0.22 in all adapted arms (naive 0.8599–0.8763; CAFT 0.8479–0.8786) and by +0.15 for frozen Geneformer (0.5921); the three-class confidence intervals did not overlap the four-class intervals, confirming that TAC rarity systematically depresses the primary endpoint. The naive-versus-CAFT comparison remained non-significant under the three-class definition, and the arm ranking was unchanged.

Out-of-corpus and cross-tissue domains bounded the collapse (Table 1, Figure 6). On the truly out-of-corpus test (GSE299152), classification fine-tuning collapsed the gradient only mildly: frozen $\rho = 0.707$ fell to 0.641 under naive fine-tuning (three-seed mean 0.641 ± 0.030), a partial rather than catastrophic loss, with a stable sign across seeds. The ordinal recipe did not transfer: ordinal-head ρ ranged from 0.35 (ordinal-naive; mean 0.350 ± 0.099) to 0.54 (ordinal-CAFT; mean 0.539 ± 0.007), below the naive arm's 0.641, and macro-F1 fell correspondingly (0.29–0.30 versus 0.452). The CORN head therefore preserves the gradient only within the pre-training corpus, where the ordinal axis is already encoded; out-of-corpus, its threshold predictions are poorly calibrated (ECE 0.56–0.61) and underperform cross-entropy.

On the cross-tissue domain (human epidermis [36], GSE147424; four ordered keratinocyte layers), the direction of the effect reversed: fine-tuning *improved* the gradient, from frozen $\rho = 0.24$ to $\rho = 0.46$ under naive fine-tuning (ordinal head 0.50). This is consistent with the headroom hypothesis: on tissue for which the pretrained representation carries no differentiation axis, cross-entropy classification learns the axis from scratch rather than overwriting an existing one. Together, the three domains delineate the boundary of TP-OAC: catastrophic within-corpus, mild out-of-corpus, and absent (inverted) cross-tissue.

Table 1: Differentiation-gradient Spearman ρ (frozen, naive cross-entropy fine-tuning, and ordinal head) across three evaluation domains. Same-corpus adapted-arm values are three-seed means with per-seed ranges in parentheses; out-of-corpus adapted-arm values are three-seed mean ± SD, with the ordinal range spanning the ordinal-naive and ordinal-CAFT arm means; frozen and cross-tissue values are single runs. Statistical control (BH-FDR and permutation nulls) applies to the same-corpus family.

Domain	Frozen	CE-naive	Ordinal (CORN)
Same-corpus (GSE153515)	0.806	−0.008 (−0.042 to +0.054)	+0.833 (0.830–0.835)
Out-of-corpus (GSE299152)	0.707	0.641 ± 0.030	0.350 ± 0.099 to 0.539 ± 0.007
Cross-tissue (epidermis)	0.24	0.46	0.50

All primary same-corpus claims survived statistical control: a single BH-FDR family of 26 comparisons (q threshold 0.05) and permutation nulls (Methods, Statistical analysis plan). Frozen and ordinal ρ were significantly positive (both $q = 0.0086$), the naive collapse was not ($q = 1.0$), and the frozen observation sat more than 120 standard deviations from the label-shuffle null.

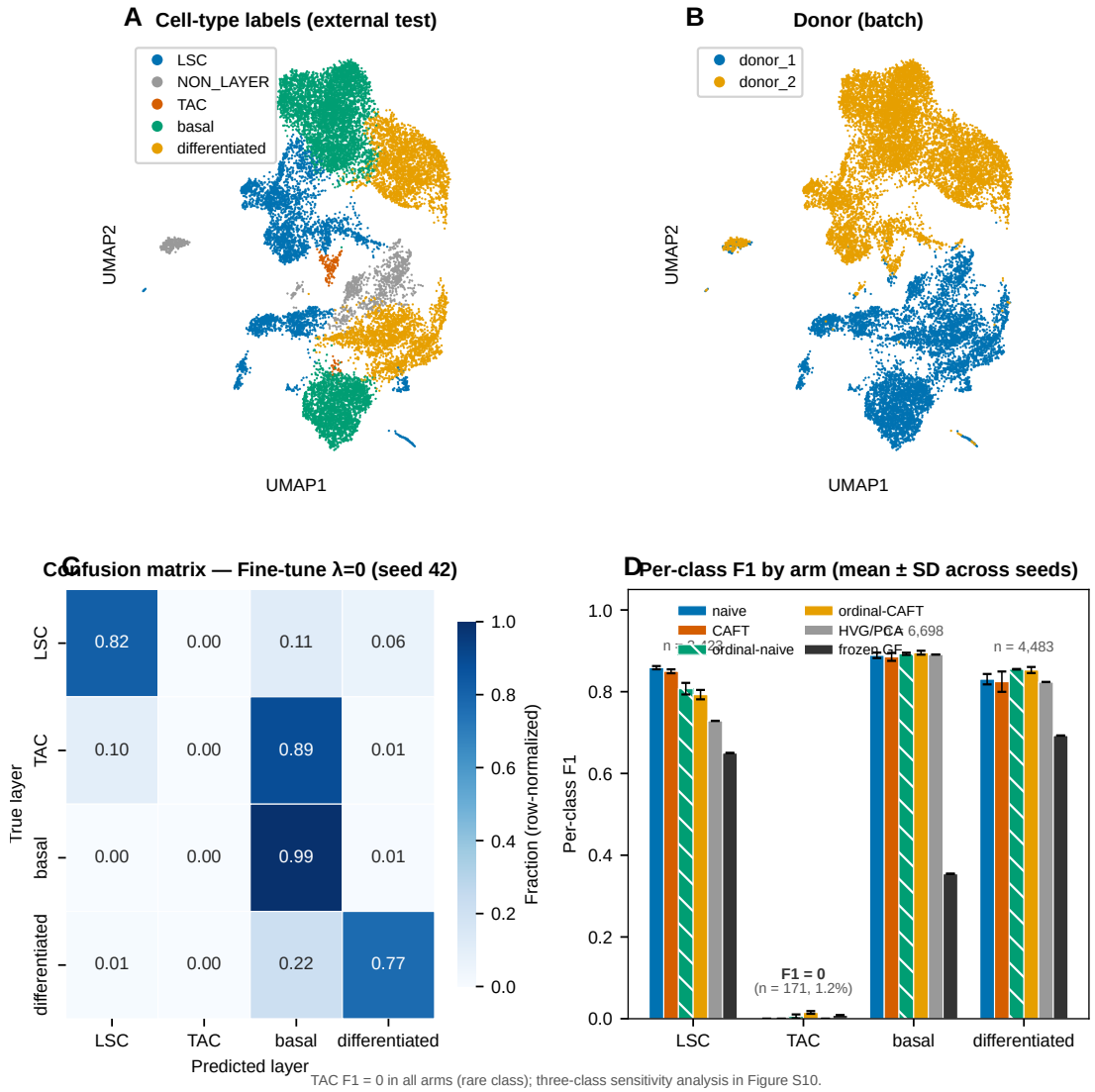


Figure 5: Same-corpus holdout: fine-tuning drives performance and collapses the gradient. External evaluation of adapted models on GSE153515 (n = 14,775 epithelial cells, two donors). (A) Cell-type UMAP (external-test labels). (B) Donor UMAP. (C) Confusion matrix. (D) Per-class F1 across six model arms (adapted arms are three-seed means; reference arms single runs); per-class support n is annotated above each class group, and the TAC column shows F1 = 0 in every adapted arm (n = 171; 1.2% of cells; three-class sensitivity analysis in Figure S10). Color key (D): blue, naive fine-tuning; vermillion, CAFT (failed control); blue-green, ordinal-naive (hatched); orange, ordinal-CAFT; gray, HVG/PCA; dark gray, frozen Geneformer. See also Figure S6.

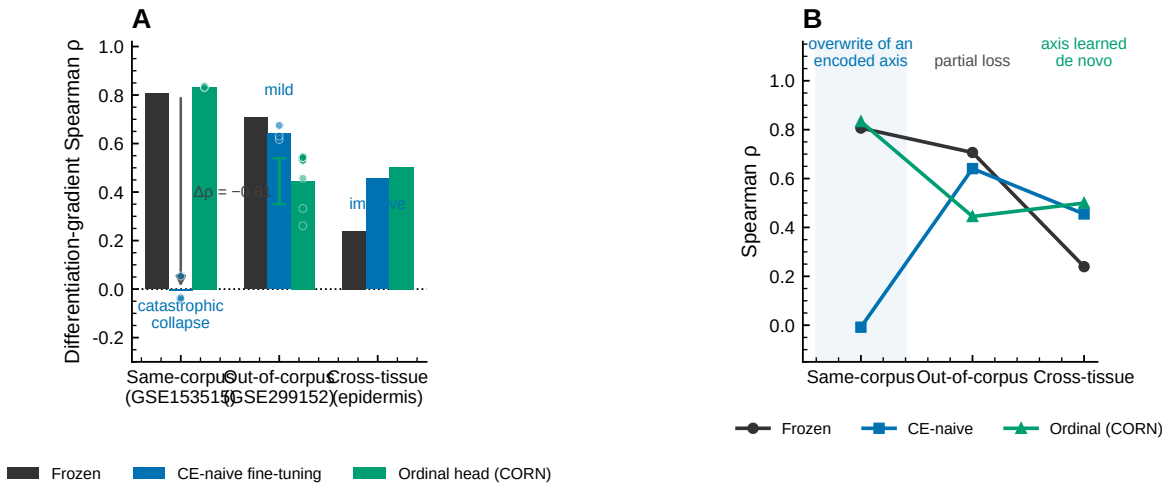


Figure 6: The collapse is domain-specific: a three-domain boundary map. (A) Spearman ρ by evaluation domain and arm (frozen, naive cross-entropy fine-tuning, ordinal head), with three-seed values shown where available and the out-of-corpus ordinal range (ordinal-naive to ordinal-CAFT) as an error bar (three-seed mean $\pm$ SD; n = 14,775 same-corpus and 21,374 out-of-corpus epithelial cells; cell counts per domain in Methods, External test datasets). Color key: dark gray, frozen Geneformer; blue, CE-naive fine-tuning; green, ordinal head (CORN; hatched). (B) Frozen-to-naive trajectories across domains: catastrophic overwrite of an encoded axis within-corpus, partial loss out-of-corpus, and de novo axis learning cross-tissue. See also Table S5.

2.3. Five gradient-preserving recipes fail: a systematic honest negative

Is the collapse avoidable? Beyond the ordinal head, we evaluated five alternative gradient-preserving recipes under the identical locked protocol (Methods, Model architectures). None preserved the differentiation gradient on the same-corpus holdout.

CAFT (continuity-regularized cross-entropy). As detailed above, the continuity objective never executed in the archived CAFT arm because of an implementation defect (Methods, Provenance and contamination audit); those checkpoints trained pure cross-entropy, their higher final-layer ρ (+0.210 versus -0.008) was run-to-run nondeterminism rather than restoration (Figure S3), and classification was indistinguishable (Δ macro-F1 = -0.004, overlapping CIs; Table S1). CAFT as implemented therefore provides no evidence that a continuity-regularized objective preserves gradient geometry.

$\mathbf{L_{cont}}$ (centroid-adjacency regularization). A correctly wired continuity term was subsequently tested in the ordinal-CAFT arm, where it executed genuinely through a registered Trainer callback (Methods). It conferred no measurable benefit beyond the ordinal head itself ($\Delta\rho = 0.005$ between ordinal-naive and ordinal-CAFT), a result isomorphic to the auxiliary-signal suppression reported for learnable-gated auxiliary objectives in frozen vision-language models [37].

AA-FT, soft-Spearman, and flow (axis-anchored and rank/flow objectives). None preserved the gradient: AA-FT $\rho = -0.0002$ ($\lambda = 0.5$) and 0.229 ($\lambda = 1.0$), soft-Spearman $\rho = -0.147$ ($\lambda = 0.5$) and -0.008 ($\lambda = 1.0$), and flow $\rho = 0.282$ ($\lambda = 0.5$), all far below the frozen reference (0.806) and the ordinal head (0.833). The failure persisted after aligning the training axis to the evaluation metric (the initial mismatch was cosine 0.470; Methods, Axis definition revision), because a single fixed axis captures at most the

centroid-trajectory direction and cannot express the full layer geometry that the metric derives per checkpoint. These arms are single-seed runs (Methods, Model architectures; per-arm values in artifacts/paper1_method_arms/).

2.4. An ordinal head parameterization preserves the differentiation gradient

The honest negative leaves open whether any fine-tuning recipe can preserve the differentiation gradient at matched accuracy. We therefore tested a constructive alternative, reparameterizing the classification head along the known ordinal axis instead of treating the four layers as unordered classes. Following the cumulative ordinal regression (CORN) formulation [38], the head emits $K-1=3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond; the loss is the sum of the three binary cross-entropies (masked to epithelial cells), and inference recovers the predicted layer as the number of thresholds with sigmoid activation ≥ 0.5. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for calibration ($P(y=k)=p_k-p_{k+1}$).

Two ordinal arms were trained under the identical locked protocol and budget: ordinal-naive ($\lambda=0$; ordinal loss only) and ordinal-CAFT ($\lambda=0.5$; ordinal loss plus the CAFT continuity term, here wired through a registered Trainer callback and therefore genuinely executed, in contrast to the archived CAFT arm). Both preserved the differentiation gradient at classification accuracy matched to the softmax arms (Table 2): three-seed mean ρ was 0.8330 (per-seed 0.8299–0.8351) for ordinal-naive and 0.8277 (0.8242–0.8321) for ordinal-CAFT, versus -0.008 for naive softmax fine-tuning, while donor-level macro-F1 was 0.6490 (95% CI 0.6100–0.6879) and 0.6479 (95% CI 0.6099–0.6860), statistically indistinguishable from the naive arm's 0.6492 (bootstrap 95% CI overlap). Layer adjacency accuracy remained $\approx$0.97 in both ordinal arms.

The near-equality of the two ordinal arms ($\Delta\rho=0.005$) indicates that the gradient preservation reflects the ordinal head parameterization itself rather than the continuity term: encoding the a priori layer order as ordered thresholds is sufficient, and the correctly wired centroid-adjacency regularization adds no measurable benefit on this axis. This is the constructive counterpart to the CAFT honest negative: a head reparameterization that respects the known ordinal structure preserves gradient geometry at matched classification performance, whereas the unordered softmax head erases it (Figure 7). This constructive result is confined to the same-corpus domain: on the truly out-of-corpus test the ordinal head preserved less gradient than cross-entropy fine-tuning (Table 1), indicating that the recipe operates by conserving an ordinal axis that pre-training has already encoded. The learned decision boundaries are themselves ordered, not merely accurate (Figure S11): the three threshold logits shift monotonically across differentiation stages, per-stage predicted probabilities align with the a priori order, and the per-cell cumulative monotonicity $P(y\geq 1)\geq P(y\geq 2)\geq P(y\geq 3)$ holds in 0.72 of cells versus 0.18 expected from an unordered head (three-seed mean 0.79; Supplementary Figure S11). Although the TAC class received no true-positive predictions in any adapted arm (F1 = 0), the ordinal head does not discard it: CORN threshold probabilities place TAC cells between LSC and basal (median predicted basal probability 0.975; Figure S11B), the biologically expected ordinal position, so the head's response to class rarity preserves rather than erases ordinal structure.

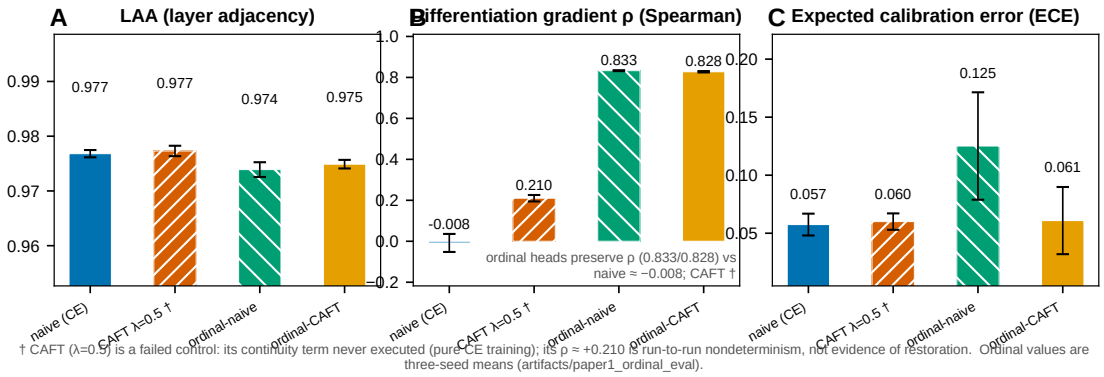


Figure 7: An ordinal head parameterization preserves the differentiation gradient. Continuity metrics under task adaptation. (A) Layer adjacency accuracy (LAA). (B) Differentiation-gradient Spearman ρ. (C) Expected calibration error (ECE). Bars denote the four adapted arms: naive (blue), CAFT (vermillion, hatched; failed control), ordinal-naive (blue-green, hatched), and ordinal-CAFT (orange) (three-seed mean ± SD; n = 14,775 cells per arm). See also Figure S3.

Table 2: External test performance by model arm (GSE153515, four classes; n = 14,775 epithelial cells from two donors). Adapted-arm values are three-seed means with donor-level bootstrap 95% CIs; reference arms are single runs (seed 42). LAA, layer adjacency accuracy; ECE, expected calibration error.

Model arm	Macro-F1	95% CI	LAA	ρ	ECE
HVG/PCA + linear	0.6192	[0.5747, 0.6637]	0.9771	0.8206	0.0117
Frozen Geneformer	0.4421	[0.3749, 0.5093]	0.9665	0.8064	0.3010
Naive FT ($\lambda = 0$)	0.6492	[0.6233, 0.6751]	0.9768	−0.0082	0.0575
CAFT ($\lambda = 0.5$)	0.6451	[0.6164, 0.6738]	0.9773	+0.2102	0.0600
Ordinal-naive ($\lambda = 0$)	0.6490	[0.6100, 0.6879]	0.9739	+0.8330	0.1251
Ordinal-CAFT ($\lambda = 0.5$)	0.6479	[0.6099, 0.6860]	0.9749	+0.8277	0.0609

3. Discussion

The central finding of this study is that differentiation geometry and classification performance are separable properties of single-cell representations. Classification fine-tuning of Geneformer-V2-104M collapsed the differentiation-gradient Spearman ρ from $\approx$0.82 (0.8206 for HVG/PCA; 0.8064 for frozen Geneformer) to -0.008 (sign-inconsistent across seeds) on the same-corpus holdout, while layer adjacency accuracy remained $\approx$0.97 in all adapted arms and macro-F1 rose to 0.6492. The continuity-regularized arm, whose objective never executed because of an implementation defect (Methods, Provenance and contamination audit), provides no evidence about continuity regularization and is reported as an honest negative; the higher final-layer ρ of its checkpoints ($+0.210$ mean) is a run-to-run effect rather than evidence of restoration. The layer-resolved analysis points to a head-proximal overwrite: the gradient signal persists in intermediate layers of the fine-tuned models, peaking at L3 with ρ of 0.81–0.84 and matching or exceeding the frozen model at the same depth (0.7656), and is selectively degraded in the layers nearest the classification head (L5–L12). The frozen model, by contrast, reaches its maximum (0.8064) precisely at the final layer, indicating that pretraining concentrates differentiation structure in the readout layer and that classification training rewrites exactly that layer. The practical implication for single-cell annotation practice is direct: pipelines that fine-tune foundation models for classification and then reuse the resulting embeddings for analyses that require continuous structure, such as trajectory inference, pseudotime ordering, or state-transition quantification, will systematically lose ordinal information even when classification accuracy is high, as our diffusion-pseudotime measurement confirms. Where continuous structure matters, intermediate-layer readouts, or frozen representations, deserve explicit consideration.

Three observations bound this collapse and its mechanism. First, the effect is domain-specific rather than universal: catastrophic on same-corpus data, mild on a truly out-of-corpus test (frozen 0.707 $\rightarrow$ naive 0.641), and reversed on cross-tissue data (frozen 0.24 $\rightarrow$ naive 0.46). The headroom hypothesis explains the cross-tissue inversion: fine-tuning destroys the gradient only where a gradient already exists to be overwritten; on tissue for which the pretrained representation carries no differentiation axis, classification supervision builds one. Second, the collapse is recipe-specific: of five gradient-preserving objectives evaluated under an identical protocol, only the ordinal head preserved the gradient, and only within-corpus; even a correctly wired continuity regularization (L_{cont}) conferred no benefit, echoing the observation that optimizers systematically suppress learnable-gated auxiliary signals in frozen models [37]. Third, single-axis recipes cannot substitute: after correcting the training-axis mismatch (cosine 0.470), axis-anchored and rank-based objectives still failed (AA-FT 0.229, soft-Spearman -0.147, flow 0.282), because a single fixed axis captures at most the centroid-trajectory direction and cannot express the full layer geometry that the evaluation metric derives per checkpoint. Together with the head-proximal overwrite described above, these observations indicate that preserving ordinal geometry under adaptation requires encoding the ordinal structure in the supervisory signal itself, at the readout where the collapse occurs.

We frame this pattern as *topology-preserving ordinal-axis collapse* (TP-OAC): supervised adaptation erases the ordinal axis of a pre-existing differentiation continuum, cluster membership and layer topology survive, with the loss graded by depth and concentrated in head-proximal layers. Three comparisons locate the term. Ordinal supervision can induce

ordered geometry, as in ordinal neural collapse (ONC) in cumulative-link regression, where class means align onto a one-dimensional axis with ordered thresholds, an ordering that holds only under fixed, equally spaced thresholds [32]; TP-OAC is the opposite failure mode, destruction of an existing ordinal axis by classification supervision, and our randomized-head ablation localizes it to the fine-tuned body. The acronym ONC also denotes an ordinal-collapse representation prior in visual navigation [39]; the phenomena are unrelated. Civale et al. characterized where differentiation signal resides in pretrained layer profiles under zero-shot conditions [33]; TP-OAC describes the change induced by fine-tuning, measured on the difference between pretrained and fine-tuned profiles rather than absolute profiles. The representational collapse R3F mitigates during fine-tuning is a generic degradation of pretrained features [40]; TP-OAC is ordinal-axis-specific and leaves discrete structure intact. To our knowledge, fine-tuning-induced collapse of an existing ordinal axis, with topology survival, has not previously been measured.

These findings connect to three parallel lines of work. First, at the level of representation geometry, TP-OAC joins the neural-collapse family of phenomena while differing from each of them: neural collapse is the terminal-phase convergence of within-class features observed during classification training [41], with analogues in neural multivariate regression [42] and in the feature distortion that fine-tuning can induce in pretrained representations [43]; unlike these within-class convergence phenomena, TP-OAC is a cross-class ordinal-axis erasure that leaves discrete layer topology intact. That full fine-tuning preferentially rewrites head-proximal layers has long been documented in NLP [44]; our contribution is its first quantitative, layer-resolved measurement on an ordinal differentiation axis in a single-cell foundation model, and the randomized-head ablation localizes the effect to the fine-tuned body rather than the readout. Related safeguards target different failure modes: R3F mitigates generic representational degradation during fine-tuning [40], and surgical fine-tuning targets out-of-distribution robustness rather than geometric fidelity [45]; scPEFT demonstrated catastrophic forgetting at the label level [23], and our measurements reveal a deeper, geometric level of forgetting in which ordinal structure is erased even though classification performance improves and discrete layer topology (LAA $\approx$0.97) survives. Objective-driven adaptation can also deliberately reorient latent geometry, as when a biological foundation model fine-tuned on perturbation profiles develops perturbation-centric axes [46]; our results document the complementary risk, in which classification supervision silently erases an ordinal axis. Second, on where differentiation information resides, Civale et al. showed that, under zero-shot conditions, differentiation-related signal concentrates at intermediate network depths rather than the final layer [33]; we add the orthogonal fine-tuning dimension, showing that task adaptation selectively overwrites exactly the readout layers while intermediate layers retain the signal; the two observations together delineate where differentiation information resides and how adaptation destroys it. The original Geneformer analysis reported that successive layers progressively separate cell types [11]; that observation concerns discrete cell-type separation, whereas TP-OAC measures the fate of the continuous ordinal axis, which collapses rather than separates. Layer-resolved dissections of pretrained models report that biological information is organized across attention layers, that residual-stream geometry is fragile under data partitioning [26, 27], and that biological knowledge exhibits multi-dimensional spectral structure [47]; our measurements complement these by quantifying the *consequence of task adaptation* on ordinal geometry rather than the internal organization

of the pretrained network. In computational pathology, the diffusion-pseudotime trajectory fidelity of foundation models predicts downstream classification [48]; we address the analogous question for single-cell differentiation hierarchies and show that the pretrained gradient, though present, is not robust to classification adaptation. Notably, zero-shot evaluations found that Geneformer's embedding space fails to retain discrete cell-type information [49], yet we measured a strong frozen ordinal gradient ($\rho = 0.8064$); ordinal geometry and discrete type identity are themselves dissociable properties of the same representation, and each must be audited separately. Third, on the evaluation of adaptation itself, our external ranking (fine-tuned > HVG/PCA > frozen) aligns with the emerging consensus from field-wide benchmarking [15–17, 19] that frozen usage is generally inadequate and that fine-tuning benefits are setting-dependent; moreover, fine-tuning can preserve pretrained gene–gene relationships when the adaptation context matches pretraining, as shown for immune-cell adaptation of CellFM [50], so collapse is not an inevitable consequence of adaptation but a property of the mismatch between the supervisory signal and the geometry to be preserved. EMP-FM reported a positive result for a dual-task objective combining classification with pseudotime-guided regression [25], whereas our CAFT is an honest negative; a plausible explanation lies in the supervisory signal: pseudotime regression provides a direct ordinal target for every cell, whereas centroid-based adjacency regularization constrains only the final representation, which, as our layer-resolved data show, is precisely the site already overwritten by the classification task. These comparisons motivate the framing we make explicit here: classification performance and representational fidelity are orthogonal evaluation dimensions. Field-wide protocols fine-tune foundation models to high annotation accuracy without auditing representation geometry [28], and benchmarking consistently reports fine-tuning gains on classification metrics [19]; yet pretrained representations designed for zero-shot use can organize developmental lineages and differentiation hierarchies without task-specific training [18], so adaptation is not inherently beneficial and its geometric cost must be measured. Encoding biological structure into the supervisory signal is one productive route: hierarchical cross-entropy losses over cell-ontology graphs improve out-of-distribution annotation by enforcing parent–child probability consistency [51], and our ordinal head extends the same principle from classification generalization to the preservation of ordinal geometry. The ranking reported here is a scoped claim about adaptation, not an assertion of foundation-model superiority over classical baselines: the frozen arm's internal-versus-external inversion ($0.8834 \rightarrow 0.4421$) marks where pretraining-only benefit ends.

As a cross-architecture probe, we evaluated the frozen representations of three further single-cell foundation models – scFoundation [52], UCE [18], and scGPT-human [13] – on both external test sets (Methods, Cross-architecture frozen probes; Table S8). All three probed models separated the four layers well (cell-level probe macro-F1: scFoundation 0.95, scGPT-human 0.85, UCE 0.79 on GSE153515), yet their ordinal arrangement along the differentiation axis differed markedly: scFoundation encoded the axis nearly as strongly as Geneformer (Spearman $\rho = 0.75$ versus 0.81 on GSE153515; 0.72 versus 0.71 on GSE299152), UCE did so intermediately (0.58; 0.71), and scGPT-human only weakly (0.22; 0.17). Both external tests are out-of-corpus for all three additional models (scGPT-human and UCE were pretrained on CELLxGENE census releases predating both tests, and GSE299152 postdates every corpus; scFoundation's 50-million-cell corpus is not publicly enumerated, so its status is inferred), so the collapse comparison has no same-corpus reference for them; the frozen-representation

measurements instead indicate that ordinal-geometry encoding is architecture-dependent rather than universal: despite uniformly high class separability, the differentiation axis is strongly organized by Geneformer and scFoundation, only partially by UCE, and barely by scGPT-human. This dissociation between class separability and ordinal arrangement extends the insight of zero-shot evaluations, which showed that fine-tuning-centric reporting conceals representational weaknesses [49], and indicates that ordinal-geometry encoding is not guaranteed across architectures.

Taken together, these results argue that classification accuracy alone is an insufficient criterion for evaluating adapted foundation models in differentiation biology: geometry-aware metrics, and layer-resolved readout in particular, expose losses that aggregate accuracy conceals. The contamination-aware paired benchmark released with this study (training atlas, a same-corpus holdout, a post-cutoff out-of-corpus test, and a locked multi-seed evaluation protocol with BH-FDR control and permutation nulls) is intended as a validation resource for such geometry-aware evaluation of future methods, in corneal biology and, we hope, beyond: evaluations in this field remain predominantly within single datasets and settings, and data splits vulnerable to leakage and coverage bias can inflate apparent performance [53]. Geometry-faithful evaluation of this kind is a prerequisite for the reliable virtual-cell representations envisioned by the AI Virtual Cell initiative [54] and complements multi-capability benchmarking efforts such as VCBench [55]. Reporting the CAFT outcome as an honest negative, rather than a promoted safeguard, is deliberate: the pre-registered promotion rule was applied without post hoc tuning, and publishing such outcomes, paired with a constructive layer-resolved diagnostic, serves the community better than selective reporting [30, 31].

3.1. Limitations of the study

This study has four principal limitations. First, it evaluates a single model family (Geneformer-V2-104M) on a single tissue (normal human corneal epithelium); the external evidence base comprises a two-donor same-corpus holdout, a three-donor post-cutoff out-of-corpus test, and a single cross-tissue epidermal dataset, and whether the dissociation between gradient geometry and layer topology, the head-proximal overwrite pattern, and the domain-specific boundary generalize across architectures, tissues, and larger donor panels remains to be tested. The cross-architecture probes (frozen scFoundation, UCE, and scGPT-human) each separated the four layers (probe macro-F1 0.79–0.95) but ranged from strong (scFoundation $\rho = 0.75$) to weak (scGPT-human $\rho = 0.22$) ordinal encoding, with all external tests out-of-corpus for these models. The domain-specific collapse map reported here is therefore measured within Geneformer-V2-104M on normal human corneal epithelium; transfer across architectures and tissues is left to dedicated evaluation. Second, the TAC class achieved F1 of 0.0 in every adapted run on GSE153515, reflecting extreme class rarity (171 of 14,775 external-test cells; 1.2%) rather than a method-specific failure, and the TAC class was absent altogether from GSE299152 (proliferation markers below 0.6% of cells); although our sensitivity analysis confirms that excluding TAC raises macro-F1 uniformly across arms without changing the ranking, conclusions regarding TAC representations are constrained by sample size in both test sets. Third, the gradient-preserving recipe arms beyond the ordinal head (AA-FT, soft-Spearman, flow) are single-seed runs: the planned three-seed expansion was gated on the seed-42 evaluation reaching a pre-specified 0.5–0.7 range, which it did not.

Their null results bound the search space for gradient-preserving adaptation, directing future recipe design toward intermediate-layer constraints and deferring multi-seed expansion until a constrained design reaches the pre-specified range. Fourth, the pre-training audit places GSE153515 within the corpus at the DOI level (53,290 cells, organ = eye; the V1 audit found zero corneal datasets), finds 10.4% of training-atlas cells overlapping corpus DOIs, and detected no cell-level overlap in bounded Signal A sampling (0.54% of the corpus). The frozen model's same-corpus performance may therefore be inflated by pre-training exposure; our conclusions rest on the relative, fine-tuning-induced measurements, which are insensitive to any shared pre-training contribution.

4. Methods

4.1. Data sources and preprocessing

Training atlas. Training and internal-test data were derived from the "Single cell atlas of the human ocular surface" collection (CELLxGENE Discover, collection identifier 0f7d022a-46c7-4e64-be4c-e34adbb78089). The data file used is the current dataset-version of CELLxGENE dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f (schema 7.1.0, published 2025-05-05), distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad (10.29 GB; MD5 39f098367af95bc3e1ee92528ceb251e; the file identifier is the dataset-version identifier at time of download, retained for traceability). The atlas is a pre-publication consortium product (CZI Cell Science / Human Cell Atlas Eye Network) without an associated journal publication or DOI, and is therefore cited as a CELLxGENE collection [9]. The collection aggregated ten constituent studies spanning corneal, limbal, scleral, and fetal ocular surface tissues [3–6, 8, 56, 57]. The aggregated AnnData object contained 120,000 cells with 20 curated cell-type labels and 31 author-assigned cell-type labels.

Cell-type labels were harmonized into a five-layer ordinal scheme reflecting the corneal epithelial differentiation continuum: LSC (including LPC and fetal epithelial progenitors; layer index 0), TAC (limbal epithelium; index 1), basal (Epi-PMC; index 2), wing (Epi-TDC/PMC; index 3), and squamous (Epi-TDC/unknown epithelium; index 4). Non-epithelial populations (immune, endothelial, stromal, conjunctival, melanocyte) were assigned layer index -1 (NON_LAYER) and excluded from ordinal analyses. For the four-class classification task evaluated on external data, the scheme was collapsed to LSC, TAC, basal, and differentiated (wing and squamous merged).

External test datasets. The study paired two external test sets with opposite pre-training-corpus status (Methods, Provenance and contamination audit). GSE153515 (GEO accession), a 10x Genomics scRNA-seq dataset of normal human limbal basal epithelium [3, 6], contained 17,587 cells from two donors (GSM4646295, GSM4646296) with an initial gene matrix of 32,738 features and is the same-corpus holdout. GSE299152 [35], a single-cell map of the ocular surface glycocalyx in adult human and mouse (human cells used here), contained 26,628 human cells from three donors with an initial gene matrix of 36,601 features and is the post-cutoff, out-of-corpus test. Disjointness from the training atlas was verified programmatically via the CELLxGENE Discover API: no overlapping GEO accession, donor identifier, or dataset hash was found between either external test and any of the ten constituent training studies; disjointness from the pre-training corpus is addressed in the contamination audit.

4.2. Fair donor-disjoint splitting protocol

To prevent donor-level information leakage between training and evaluation, a strict donor-disjoint split was applied to the training atlas prior to any model fitting. Donors were partitioned using a deterministic random assignment (seed = 42), yielding 49 donors (99,606 cells) for training and 25 donors (20,394 cells) for internal testing, with zero donor overlap between partitions. The split protocol ensured representation of all 20 curated labels and all 31 author-assigned labels in both partitions. The split manifest, including per-partition cell counts, donor lists, label distributions, and SHA-256 checksums of the resulting train and test AnnData files, was locked and archived (artifacts/paper1_fair_split/manifest.json). No cells were reassigned after manifest generation.

4.3. External test annotation pipeline

Because GSE153515 was deposited with an empty observation metadata table, a fully reproducible de novo annotation pipeline was implemented (scripts/protocol_15_annotate_gse153515.py): (1) donor assignment from the 10x barcode suffix convention; (2) marker scoring on the full 32,738-gene matrix using eight curated marker gene sets spanning LSC, TAC, basal, differentiated, conjunctival, melanocyte, immune, and stromal identities; (3) quality-control filtering by minimum gene count and maximum mitochondrial fraction (15,827 of 17,587 cells retained, 89.9%, spanning 10,605 genes); (4) integration and clustering: highly variable gene selection, log-normalization, scaling, PCA, Harmony batch correction (n_pcs = 30) [58], and Leiden clustering (resolution 0.6) yielding 16 clusters; (5) DEG-based cell-type assignment via Wilcoxon rank-sum test with an enrichment score $(n_{\text{in top200}}/n_{\text{total}}) \times (1 - \text{mean_rank}/200)$ against the top 200 cluster-specific DEGs, with clusters below an epithelial enrichment threshold of 0.3 classified as NON_LAYER; and (6) harmonization to the four-class scheme (basal 6,698; differentiated 4,483; LSC 3,423; TAC 171 cells) with non-epithelial cells excluded. All annotation parameters were locked in a configuration file (config/external_test_gse153515.yml) before any model evaluation. The second external test, GSE299152, was annotated with a mirrored pipeline (scripts/protocol_15b_annotate_gse299152.py; config/external_test_gse299152.yml) with two recorded deviations: Endothelial and Stromal marker sets were added to the NON_LAYER gate, and Leiden resolution was auto-tuned to 0.7 (15 clusters). After QC (23,970 of 26,628 cells retained, 90.0%; 11,274 genes), harmonization yielded LSC 15,921, basal 1,099, and differentiated 4,354 cells, with an empty TAC class (proliferation markers below 0.6% of cells) and 2,596 non-epithelial cells excluded; low-confidence clusters (melanocyte enrichment 0.199 below the 0.3 gate; MT-high and stress clusters with near-zero enrichment) were default-assigned to LSC/LPC and flagged in the annotation report.

4.4. Model architectures

Four model configurations were evaluated under an identical experimental protocol (same preprocessing, data split, classification head, training epochs, and random seeds); two additional ordinal-head arms (Arms 5–6) and five gradient-preserving objective arms (Arms 7–11) shared the same protocol and budget:

Arm 1—HVG/PCA + linear classifier (transparent baseline). Highly variable genes were selected from the training partition; expression values were log-normalized and

scaled, PCA was applied, and a linear classifier was trained on the resulting principal components.

Arm 2—Frozen Geneformer + linear probe. Geneformer-V2-104M (BertForMaskedLM architecture; hidden dimension 768, 12 layers, 12 attention heads, vocabulary 20,275) was loaded from the Hugging Face model hub (snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5; model weights SHA-256 fff5cba2...), pretrained on Genecorpus-104M (∼104 million cells from 2,903 datasets) [11, 12]. All Transformer parameters were frozen; only a linear classification head appended to the [CLS] token representation was trained.

Arm 3—Geneformer naive fine-tuning ($\lambda = 0$). The full model was fine-tuned end-to-end with a standard cross-entropy classification loss over the 31 author-assigned cell-type classes (no continuity regularization), serving as the fair adaptation comparator; predictions were merged to the four epithelial differentiation classes for external evaluation.

Arm 4—Geneformer + CAFT ($\lambda = 0.5$). The full model was fine-tuned with the CAFT composite loss (Methods, CAFT loss formulation) at a single fixed regularization strength, testing CAFT as a candidate safeguard for ordered differentiation geometry.

Arm 5—Geneformer + ordinal-naive head ($\lambda = 0$). The full model was fine-tuned end-to-end with the CORN ordinal loss (Methods, Ordinal head formulation) and no continuity regularization.

Arm 6—Geneformer + ordinal-CAFT head ($\lambda = 0.5$). The full model was fine-tuned with the CORN ordinal loss plus the CAFT continuity term, wired through a registered Trainer callback so that the continuity objective was genuinely executed (in contrast to the archived CAFT arm).

Arms 7–11—axis-anchored, soft-Spearman, and flow objectives. Three further gradient-preserving objectives were evaluated with a four-class cross-entropy head on the revised centroid-PCA differentiation axis (Methods, Axis definition revision): Arm 7 (AA-FT, $\lambda = 0.5$) and Arm 8 (AA-FT, $\lambda = 1.0$) augment the classification loss with a projection-constraint term pulling embeddings onto the fixed differentiation axis [59]; Arm 9 (soft-Spearman, $\lambda = 0.5$) and Arm 10 (soft-Spearman, $\lambda = 1.0$) add a differentiable Spearman surrogate (soft rank via sigmoid pairwise comparisons with a diagonal correction, ABAG-Rank formulation [60], with a margin surrogate [61]); and Arm 11 (flow, $\lambda = 0.5$) adds a flow-consistency objective constraining embeddings toward ordered transitions between adjacent layers. These arms used the same preprocessing, data split, and training budget as Arms 3–6.

Arms 3–6 were each trained with three independent random seeds (42, 123, 456) for three epochs. Arms 7–11 were trained with seed 42 only: a three-seed expansion was gated on the seed-42 evaluation reaching a pre-specified differentiation-gradient range (0.5–0.7) on the same-corpus holdout, which no arm reached (Results). Training used a per-device batch size of 8 with gradient accumulation of 4 (effective batch size 32), bfloat16 mixed precision, gradient checkpointing, linear learning-rate warmup over 500 steps, weight decay 0.001, and learning rate 5×10^{-5}.

4.5. CAFT loss formulation

The CAFT objective augmented standard cross-entropy classification with a continuity regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{cont}}, \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy classification loss over the 31 author-assigned cell-type classes of the training atlas (merged to the four epithelial differentiation classes for external evaluation) and $\mathcal{L}_{\text{cont}}$ penalizes deviation from expected adjacency structure. For each training cell x_i with assigned differentiation layer $\ell_i \in \{0, 1, 2, 3, 4\}$, the continuity loss was

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{\ell' \in \mathcal{A}(\ell_i)} \cos\Big(\mathbf{z}_i,\ \text{sg}(\mathbf{c}_{\ell'})\Big), \tag{2}$$

where $\mathbf{z}_i$ is the ℓ_2-normalized [CLS] embedding of cell x_i, $\mathcal{A}(\ell_i)$ denotes the adjacent layers ($\ell_i \pm 1$, clipped to $[0, 4]$), $\mathbf{c}_{\ell'}$ is the centroid of training embeddings assigned to layer ℓ', and $\text{sg}(\cdot)$ is the stop-gradient operator. Layer centroids were precomputed at the start of each epoch from detached embeddings and held fixed throughout that epoch. NON_LAYER cells ($\ell = -1$) were excluded from the continuity term but contributed to $\mathcal{L}_{\text{cls}}$.

4.6. Ordinal head formulation

To test whether the collapse of the differentiation gradient is caused by treating the four ordered layers as unordered softmax classes, Arms 5–6 replaced the softmax classification head with a cumulative ordinal head following the CORN formulation [38]. For $K = 4$ ordinal classes, the head emits $K - 1 = 3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond ($P(y \geq k)$); the loss is the sum of the three binary cross-entropies, masked to epithelial cells:

$$\mathcal{L}_{\text{ord}} = \sum_{k=1}^{K-1} \text{BCE}\Big(\sigma(f_k(\mathbf{z})),\ [y \geq k]\Big), \tag{3}$$

where $f_k(\mathbf{z})$ is the k-th threshold logit on the [CLS] representation $\mathbf{z}$, σ is the sigmoid, and $[y \geq k]$ is the binary target indicating that the cell's ordinal layer is at least k. At inference the predicted class is the number of thresholds with $\sigma(f_k(\mathbf{z})) \geq 0.5$. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for calibration ($P(y = k) = p_k - p_{k+1}$, with $p_0 = 1$, $p_K = 0$). The ordinal-CAFT arm (Arm 6) added the CAFT continuity term ($\mathcal{L} = \mathcal{L}_{\text{ord}} + \lambda \cdot \mathcal{L}_{\text{cont}}$), with $\mathcal{L}_{\text{cont}}$ wired through a registered `TrainerCallback` so that the continuity objective was genuinely executed at each epoch start, in contrast to the archived CAFT arm, where the equivalent hook was never dispatched.

4.7. Evaluation metrics

All metrics were computed on the locked external test sets using the Tier-B evaluation pipeline (corneal_fm/evaluation.py).

Primary endpoint. Donor-level macro-averaged F1 (macro-F1) on the four-class external test set, with 95% confidence intervals estimated by nonparametric bootstrap resampling of donors (n_boot = 1,000; seed = 42). Per-class precision, recall, F1, and support were reported for each arm and seed.

Secondary endpoints. (i) Differentiation-gradient rank correlation (Spearman ρ): an ordinal-structure metric computed solely from the layer-centroid geometry of the learned embeddings, without marker-gene expression. For each of the four layer classes (NON_LAYER excluded), the centroid of the cell embeddings was computed; the principal differentiation axis was defined as the first singular vector of the centered layer centroids, oriented so that pseudotime increases from LSC to differentiated; each cell was assigned a pseudotime equal to the projection of its embedding onto this axis; and ρ was the Spearman correlation between per-cell pseudotime and the a priori layer ordinal. (ii) Layer Adjacency Accuracy (LAA): the fraction of nearest-neighbor pairs in the learned embedding space belonging to adjacent or identical differentiation layers. (iii) Expected Calibration Error (ECE): the weighted mean absolute difference between predicted confidence and observed accuracy across probability bins. (iv) Internal silhouette coefficient and gap statistic were computed as diagnostic-only measures and were not used for model selection or promotion decisions.

Layer-resolved gradient measurement. At inference, all thirteen hidden states (embedding output L0 and layers L1–L12) were collected for each of the 14,775 external-test epithelial cells, and the differentiation-gradient ρ was recomputed at each depth from the position-0 [CLS] representation (scripts/measure_layerwise_rho.py). A sanity gate required the L12 ρ of every model to reproduce the corresponding single-value measurement within 10^{-6} (all seven models passed bit-exactly). At L0 the position-0 [CLS] representation is constant across cells, so ρ is undefined (NaN) and was excluded (expected behavior). For the randomized-head ablation, the trained classification head was replaced with a freshly initialized, untrained head and hidden states were collected without further training; the maximum hidden-state difference from the intact model was 0 for both the fine-tuned and frozen bodies. Cell-level bootstrap 95% confidence intervals at L12 were computed with n_boot = 1,000 resamples (seed = 42; scripts/bootstrap_layerwise_rho.py).

Trajectory inference. Diffusion pseudotime (scanpy DPT) was computed on PCA-reduced external embeddings (n_pcs = 30, n_neighbors = 30, n_dcs = 10), rooted at the LSC class, and correlated (Spearman) with the reference layer-ordering pseudotime for each arm (scripts/analyze_trajectory_continuity.py).

TAC sensitivity analysis. Because the TAC class is extremely rare (171/14,775 cells; 1.2%), donor-level macro-F1 was recomputed excluding TAC (three classes), with the same bootstrap procedure, to assess whether the rare class drives the primary endpoint (scripts/analyze_tac_sensitivity.py).

4.8. Axis definition revision (protocol deviation)

The axis-anchored and soft-Spearman arms (Arms 7–10) were initially trained against a marker-score differentiation axis. An evaluation of seed 42 exposed a training-axis versus evaluation-metric direction mismatch: the marker-score axis and the per-layer-centroid-PCA axis used by the evaluation metric (Secondary endpoints) aligned with cosine 0.470. On 2026-08-10 the axis definition was revised (protocol deviation, user-approved; changelog 2026-08-10-paper1-axis-refit-option-a) to a per-layer-centroid-PCA axis computed over all epithelial training cells, bit-identical to the evaluation metric's axis (corneal_fm/axis_ref.py, mode = "centroid_pca"; cache schema v2); the marker-score axis is retained as a reference mode. Arms 7–11 and the cross-tissue evaluation used the revised axis.

4.9. Cross-architecture frozen probes (scGPT-human, scFoundation, UCE)

To probe whether the ordinal-geometry measurements generalize across single-cell foundation models without incurring a second fine-tuning protocol, the frozen embeddings of scGPT-human (perturblab/scgpt-human checkpoint; 12-layer Transformer, 512-dimensional hidden state, continuous log1p input, CLS cell embedding, 51.86M parameters; scgpt 0.2.5, git commit cebd6fae) were evaluated on both external test sets. Cells were tokenized with the model's 60,697-gene vocabulary (append_cls = True, maximum sequence length 1,200 with the CLS token kept first) and embedded at the CLS position of the final layer; the differentiation-gradient ρ was computed with the same marker-gradient metric as all other arms, and class separability was measured with a four-class logistic-regression probe (cell-level 80/20 split, five repeats, mean macro-F1). Gene-symbol mapping for GSE153515 was verified against HGNC: 10,379 of 10,605 ENSEMBL identifiers resolved authoritatively, of which 32 of the 9,175 vocabulary-matched genes (0.35%) carry post-download HGNC renaming (none core corneal markers), with no effect on the input.

Frozen scFoundation (01B-resolution checkpoint; genbio-ai/scFoundation Hugging Face mirror, checkpoint sha256 9f40bf32…; xTrimoGene encoder–decoder, 119.25M parameters [52]) was embedded with the official inference pipeline (get_embedding.py; input_type singlecell, output_type cell, pool_type all, tgthighres a5, pre_normalized F, version ce), yielding per-cell embeddings of dimension 3,072 (concatenation of last-layer, second-to-last, max-pooled, and mean-pooled hidden states over the 19,264-gene vocabulary). Raw counts were aligned to the model's 19,264 HGNC-symbol vocabulary (9,162 of 9,175 input genes matched for GSE153515; 10,296 of 11,274 for GSE299152; unmatched genes zero-filled). scFoundation's pretraining corpus (over 50 million human cells) is not publicly enumerated, so corpus membership of the two test sets could not be verified and is reported as inferred out-of-corpus.

Frozen UCE (Universal Cell Embedding; 33-layer checkpoint 33l_8ep_1024t_1280.torch, Figshare 10.6084/m9.figshare.24320806, checkpoint sha256 3f458726…; [18]) cell embeddings were computed with the official inference script (eval_single_anndata.py; species human, sample_size 1,024 expression-weighted genes sampled per cell with replacement, pad_length 1,536, CLS-token hidden state of dimension 1,280, L2-normalized). Because UCE cell representations depend on stochastic gene sampling, each dataset was embedded three times (seeds 42, 123, 456) and the differentiation-gradient ρ and probe macro-F1 were averaged across repeats (per-repeat values in Table S8). UCE was pretrained on the CELLxGENE census 2023-04-25 release; GSE153515 is absent from the closest queryable census release (2023-05-15) and GSE299152 postdates the corpus, so both test sets are reported as out-of-corpus (inference).

4.10. CAFT promotion criteria

CAFT (Arm 4) was promoted over naive fine-tuning (Arm 3) only if all four criteria were satisfied simultaneously: (1) external continuity improvement (ρ or LAA) relative to Arm 3; (2) non-inferior classification (bootstrap 95% CI overlap or non-inferiority within a prespecified margin); (3) seed and donor consistency of the improvement direction; and (4) survival of the locked provenance audit and donor- and study-level disjointness verification. Failure of any single criterion retained Arm 3 as the preferred strategy, with CAFT reported as an unsuccessful candidate; no post hoc tuning or selective reporting was permitted.

4.11. Provenance and contamination audit

All computational artifacts were version-locked prior to analysis: model weights (retrieved 2026-07-23), training and test data files (downloaded 2026-07-22), and SHA-256 checksums for model weights and train/test AnnData were recorded, and the fair-split manifest was generated before any training. Because Geneformer was pretrained on Genecorpus-104M, we assessed whether corneal or ocular surface datasets from our training atlas might have been included in the pretraining corpus. For Genecorpus-30M (V1; 561 datasets with a public manifest), all 43 eye-related datasets were identified; all were retinal or retinal pigment epithelium datasets, with zero corneal datasets, and all eight GEO accessions corresponding to our training studies returned no matches. For Genecorpus-104M (V2; 2,903 datasets), the official corpus source table (Supplementary table MOESM3, sheet `genecorpus_104m`; 563 unique DOIs) was downloaded from the Nature Computational Science supplementary materials and used for a DOI-level audit. GSE153515's primary publication DOI (10.1016/j.jtos.2020.12.004; Li et al., The Ocular Surface 2021) resolves to a row in the corpus table (53,290 cells, organ = eye); because all corpus datasets predate assembly and GSE153515 was public from 2021-01-14, no temporal exemption applies, and GSE153515 is reported as same-corpus (least conservative). GSE299152 [35] (GEO public 2025-11-18) postdates corpus assembly and its DOI is absent from the table, making it the post-cutoff, out-of-corpus (most conservative) test. For the cross-architecture probe, scGPT-human was pretrained on the CELLxGENE census 2023-05-15 release (33,758,887 unique normal human cells; [13] Data Availability). GSE153515 is absent from both the CELLxGENE curation catalog and the census 2023-05-15 dataset manifest (verified via the CZ curation API and the census S3 h5ads directory), and GSE299152 (GEO released 2025-11-18) postdates the corpus; both external tests are therefore out-of-corpus for scGPT. A secondary finding: two training-atlas studies (lako_GSE155683, 9,972 cells; li_GSE147979, 423 cells) resolve to corpus DOIs (10,395 of 99,606 cells, 10.4%), disclosed for completeness; this does not affect the external-test conclusions, which rest on donor- and study-level disjointness from the training atlas. As a corroborating cell-level check in the spirit of scContam [62], bounded Signal A fingerprint sampling (0.54% of the corpus, 491,227 fingerprints; vectorized MinHash m = 128, b = 16, r = 8, banded LSH, exact Jaccard confirmation) found zero hits for the 15,827 GSE153515 cells (max Jaccard 0.0), with 100% positive-control recall and zero negative-control false positives; because full-corpus coverage (approximately 600 GB) was not attained, this run validates the pipeline but does not constitute exclusion evidence. The DOI-level audit is the primary evidence.

CAFT implementation audit. A post hoc audit of the six archived fine-tuning runs (the archived copies of scripts/caft_gf.py in the six save directories, byte-identical to the current script; MD5 8241ff23dea92ff1789e23c569108f2a) established that the continuity term $\mathcal{L}_{\text{cont}}$ never executed in any run. In Transformers 4.57, the `on_epoch_begin` hook is dispatched only to registered `TrainerCallback` objects; `CAFTTrainer.on_epoch_begin` is a plain method that no callback ever invokes, so `layer_centroids` was never populated and `compute_loss` always fell back to $\mathcal{L}_{\text{cls}}$. The training history of every run contains only the plain `loss` key, with no `L_cont` or `caft_loss` entries. The $\lambda = 0.5$ checkpoints therefore trained pure cross-entropy, identical in objective to the $\lambda = 0$ arm, and the separation between the two groups' final-layer ρ (naive mean -0.008; $\lambda = 0.5$ mean $+0.210$) is attributable to run-to-run nondeterminism, bfloat16 training over 7,473 steps and stochastic data preparation that permutes class identifiers and row order between runs, rather than to the CAFT objective.

The locked results_*.json historical values are retained unchanged; ECE values recomputed with the corrected half-open binning convention are recorded in artifacts/paper1_pipeline_audit/ece_corrected.json. A correctly wired continuity objective was subsequently tested in the ordinal-CAFT arm (Methods, Ordinal head formulation, Arm 6; Results, ordinal head), where it executed genuinely but conferred no measurable gradient benefit beyond the ordinal head itself; the axis-anchored, soft-Spearman, and flow arms (Arms 7–11), trained on the revised centroid-PCA axis, all failed to preserve the gradient and are reported as additional honest negatives (Results, five gradient-preserving recipes fail).

4.12. Statistical analysis plan

All statistical analyses were prespecified in the locked evaluation configuration prior to model training. The primary comparison was between Arm 3 and Arm 4, with Arms 1 and 2 as reference baselines. All primary same-corpus claims were submitted to a single Benjamini–Hochberg FDR family of 26 comparisons (q threshold 0.05), spanning, per arm, gradient-positivity ($\rho > 0$), layer-adjacency-above-chance (LAA > chance), and classification-parity (F1 = naive F1) tests across the frozen, naive, CAFT, ordinal, AA-FT, soft-Spearman, and flow arms (scripts/fdr_nulls_analysis.py; artifacts/paper1_fdr_nulls/bh_fdr_table.json). Permutation nulls were constructed by label shuffling (500 replicates) and feature shuffling (300 replicates) of the differentiation-gradient measurement, and observed ρ values were standardized against the null mean and SD (z-scores; artifacts/paper1_fdr_nulls/null_controls.json). Uncertainty was quantified by nonparametric bootstrap resampling at the donor level (n_boot = 1,000; seed = 42). Model stability was assessed across three independent training seeds, reported as mean ± range. All hypothesis tests were two-sided; $p < 0.05$ was considered significant. No interim analyses or adaptive stopping rules were employed.

4.13. Software and reproducibility

All analyses were performed in Python 3.12, using Scanpy [63], harmonypy [58, 64], Hugging Face Transformers [65], PyTorch [66], and scikit-learn [67]. The CAFT training pipeline was implemented as a custom extension of the Hugging Face Trainer class (corneal_fm/caft.py); evaluation used the Tier-B module (corneal_fm/evaluation.py); external annotation used the locked pipeline scripts and configuration files. Gradient-preserving objective training used corneal_fm/axis_ref.py and scripts/method_arm_gf.py; the three-domain boundary evaluation used scripts/run_method_arms_eval.py, scripts/epidermis_tpoac_eval.py, and scripts/summarize_tpoac_boundary.py; FDR and permutation-null control used scripts/fdr_nulls_analysis.py.

Resource availability

Lead contact. Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Qi Dai (dq@mail.eye.ac.cn).

Materials availability. This study did not generate new unique reagents or biological materials; all data are public (below).

Data and code availability.

- **Training data.** CELLxGENE Discover collection 0f7d022a-46c7-4e64-be4c-e34adbb78089 (Single cell atlas of the human ocular surface; Human Ocular Surface Atlas), current dataset-version of dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f, distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad; MD5 39f098367af95bc3e1ee92528ceb251e.

- **External test data.** GEO GSE153515 [3]; 17,587 cells from two donors (GSM4646295, GSM4646296). GEO GSE299152 [35]; 21,374 epithelial cells from three donors after QC. Cross-tissue: GEO GSE147424 [36].

- **Cross-architecture probes.** scGPT-human checkpoint (perturblab/scgpt-human; scgpt 0.2.5, git commit cebd6fae); scFoundation 01B-resolution checkpoint (genbio-ai/scFoundation Hugging Face mirror); UCE 33-layer checkpoint (Figshare 10.6084/m9.figshare.24320806); frozen-probe diagnostics in artifacts/scgpt_repro/frozen_diag/, artifacts/scfoundation_repro/frozen_diag/, and artifacts/uce_repro/frozen_diag/.

- **Model.** Geneformer-V2-104M, Hugging Face model hub (ctheodoris/Geneformer; snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5).

- **Code.** Analysis code is available at `https://github.com/Xu-Kunk/Cornea-Ai` and will be made publicly accessible upon acceptance.

- **Evaluation scripts.** scripts/run_tier_b_eval.py; scripts/run_tier_b_baselines.py; scripts/measure_layerwise_rho.py; scripts/bootstrap_layerwise_rho.py; scripts/analyze_trajectory_continuity.py; scripts/analyze_tac_sensitivity.py; scripts/protocol_15b_annotate_gse299152.py; scripts/method_arm_gf.py; scripts/run_method_arms_eval.py; scripts/epidermis_tpoac_eval.py; scripts/summarize_tpoac_boundary.py; scripts/fdr_nulls_analysis.py.

- **Results.** artifacts/paper1_tier_b_eval/*.json (six fine-tuned-arm and two baseline-arm result files); artifacts/paper1_tier_b_eval_gse299152/*.json (out-of-corpus results); artifacts/paper1_ordinal_eval/*.json (ordinal arms); artifacts/paper1_epidermis_eval/epidermis_rho_results.json (cross-tissue); artifacts/paper1_fdr_nulls/ (bh_fdr_table.json, null_controls.json, contamination_audit_protocol.md).

- **Reproducibility.** All experiments are reproducible with fixed random seeds (42, 123, 456), locked configuration files (config/external_test_gse153515.yml, config/external_test_gse299152.yml), and the SHA-256-locked fair-split manifest.

References

[1] G. Cotsarelis, S.-Z. Cheng, G. Dong, T.-T. Sun, R. M. Lavker, Existence of slow-cycling limbal epithelial basal cells that can be preferentially stimulated to proliferate: implications on epithelial stem cells, Cell 57 (2) (1989) 201–209. doi:10.1016/0092-8674(89)90958-6.

[2] H. S. Dua, A. Azuara-Blanco, Limbal stem cells of the corneal epithelium, Survey of Ophthalmology 44 (5) (2000) 415–425. doi:10.1016/S0039-6257(00)00109-0.

[3] D.-Q. Li, S. Kim, J.-M. Li, Q. Gao, J. Choi, F. Bian, J. Hu, Y. Zhang, R. Lu, Y. Lu, S. C. Pflugfelder, Single-cell transcriptomics identifies limbal stem cell population and cell types mapping its differentiation trajectory in limbal basal epithelium of human cornea, The Ocular Surface 20 (2021) 20–32. doi:10.1016/j.jtos.2020.12.004.

[4] J. Collin, R. Queen, D. Zerti, B. Dorgau, M. Georgiou, I. Djidrovski, R. Hussain, J. M. Coxhead, A. Joseph, P. Rooney, S. Lisgo, F. Figueiredo, L. Armstrong, M. Lako, A single cell atlas of human cornea that defines its development, limbal progenitor cells and their interactions with the immune cells, The Ocular Surface 21 (2021) 279–298. doi:10.1016/j.jtos.2021.03.010.

[5] P. Català, N. Groen, J. A. Dehnen, J. S. Soares, A. J. H. van Velthoven, R. M. M. A. Nuijts, M. M. Dickman, V. L. S. LaPointe, Single cell transcriptomics reveals the heterogeneity of the human cornea to identify novel markers of the limbus and stroma, Scientific Reports 11 (1) (2021) 21727. doi:10.1038/s41598-021-01015-w.

[6] J.-M. Li, S. Kim, D.-Q. Li, Q. Gao, J. Choi, F. Bian, J. Hu, Y. Zhang, R. Lu, S. C. Pflugfelder, Single-cell transcriptomics identifies a unique entity and signature markers of transit-amplifying cells in human corneal limbus, Investigative Ophthalmology & Visual Science 62 (9) (2021) 36. doi:10.1167/iovs.62.9.36.

[7] A. J. Ligocki, et al., Molecular characteristics and spatial distribution of adult human corneal cell subtypes, Scientific Reports 11 (2021) 16323. doi:10.1038/s41598-021-94933-8.

[8] G. Maiti, M. Alameen, D. Nayak, M. S. Sridhar, S. Ghosh, Single cell rna-seq of human cornea organoids identifies cell fates of a developing immature cornea, PNAS Nexus 1 (5) (2022) pgac246. doi:10.1093/pnasnexus/pgac246.

[9] R. Chen, CZI Cell Science / Human Cell Atlas Eye Network, Single-cell atlas of the human ocular surface, CELLxGENE Discover collection 0f7d022a-46c7-4e64-be4c-e34adbb78089, accessed 2026. Pre-publication consortium atlas; cited as a collection (2026).
URL https://cellxgene.cziscience.com/collections/0f7d022a-46c7-4e64-be4c-e34adbb78089

[10] J. A. Arts, S. Fallo, M. S. Florencio, J. G. A. Smits, D. Lima Cunha, J. A. Y. Roubroeks, et al., Prediction of cell states and key transcription factors of the human cornea through integrated single-cell omics analyses, PNAS Nexus 4 (8) (2025) pgaf235. doi:10.1093/pnasnexus/pgaf235.

[11] C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin, Z. R. Al Sayed, M. C. Hill, H. Mantineo, E. M. Brydon, Z. Zeng, X. S. Liu, P. T. Ellinor, Transfer learning enables predictions in network biology, Nature 618 (7965) (2023) 616–624. doi:10.1038/s41586-023-06139-9.

[12] H. Chen, et al., Scaling and quantization of large-scale foundation model enables resource-efficient predictions in network biology, Nature Computational Science 6 (5) (2026) 450–463. doi:10.1038/s43588-026-00972-4.

[13] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, B. Wang, scgpt: toward building a foundation model for single-cell multi-omics using generative ai, Nature Methods 21 (8) (2024) 1470–1480. doi:10.1038/s41592-024-02201-0.

[14] F. Yang, W. Wang, F. Wang, Y. Fang, D. Tang, J. Huang, H. Lu, J. Yao, scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data, Nature Machine Intelligence 4 (10) (2022) 852–866. doi:10.1038/s42256-022-00534-z.

[15] S. Hou, P. Yang, W. Ma, J.-X. Wang, X. Zhou, A unified framework enables accessible deployment and comprehensive benchmarking of single-cell foundation models, bioRxiv preprint (2026). doi:10.64898/2026.01.06.698060.

[16] J. Wu, Q. Ye, Y. Wang, et al., Biology-driven insights into the power of single-cell foundation models, Genome Biology 26 (2025) 334. doi:10.1186/s13059-025-03781-6.

[17] T. Liu, K. Li, Y. Wang, H. Li, H. Zhao, Evaluating the utilities of foundation models in single-cell data analysis, Advanced Science 13 (27) (2026) e14490. doi:10.1002/advs.202514490.

[18] Y. Rosen, Y. Roohani, A. Agrawal, L. Samotorčan, Tabula Sapiens Consortium, S. R. Quake, J. Leskovec, Universal cell embedding provides a foundation model for cell biology, Nature 656 (8126) (2026) 183–191. doi:10.1038/s41586-026-10689-z.

[19] P. Qiu, Q. Chen, H. Qin, S. Fang, Y. Zhang, Y. Zhang, et al., Biollm: A standardized framework for integrating and benchmarking single-cell foundation models, Patterns 6 (8) (2025) 101326. doi:10.1016/j.patter.2025.101326.

[20] T. Liu, T. Chen, W. Zheng, X. Luo, Y. Chen, H. Zhao, Embeddings from language models are good learners for single-cell data analysis, Patterns 7 (2) (2026) 101431. doi:10.1016/j.patter.2025.101431.

[21] C. Wang, Y. T. Chen, et al., Evaluating large language models for evidence-based clinical question answering, Patterns 7 (5) (2026) 101519. doi:10.1016/j.patter.2026.101519.

[22] Z. Ma, J. Li, J. Hu, B. Yu, Uncovering smooth structures in single-cell data with neighbor embeddings guided by predictability-computability-stability, Patterns 7 (7) (2026) 101621. doi:10.1016/j.patter.2026.101621.

[23] F. He, R. Fei, J. E. Krull, Y. Yu, X. Zhang, X. Wang, et al., Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scpeft, Nature Machine Intelligence 8 (2026) 118–133. doi:10.1038/s42256-025-01170-z.

[24] X. Zhou, Z. Wang, Y. Ling, Q. Tian, Z. Zhang, Y. Li, L. Chen, P. Zhou, Benchmarking zero-shot single-cell foundation model embeddings for cellular dynamics reconstruction, bioRxiv preprint (2026). doi:10.64898/2026.03.10.710748.

[25] S. Pan, E. Withnell, C. Celik, M. Secrier, From cell types to cell states: adapting foundation models for cellular plasticity, bioRxiv preprint, v2 January 2026 (2024). doi:10.1101/2024.08.16.608311.

[26] I. Kendiukhov, et al., Attention layers in single-cell foundation models organize as protein–protein interactions in early layers and transcriptional regulation in late layers, BMC Genomics 27 (2026) 12965. doi:10.1186/s12864-026-12965-8.

[27] I. Kendiukhov, et al., Residual-stream geometry in single-cell foundation models, BMC Bioinformatics 27 (2026) 6538. doi:10.1186/s12859-026-06538-5.

[28] S. Ding, J. Li, R. Luo, H. Cui, B. Wang, R. Chen, scgpt: end-to-end protocol for fine-tuned retinal cell type annotation, Nature Protocols 21 (3) (2026) 873–893. doi:10.1038/s41596-025-01220-1.

[29] M. Teles, et al., Unraveling learning characteristics of transformer models for molecular design, Patterns 6 (12) (2025) 101392. doi:10.1016/j.patter.2025.101392.

[30] W. A. Van Amsterdam, et al., When accurate prediction models yield harmful self-fulfilling prophecies, Patterns 6 (4) (2025) 101229. doi:10.1016/j.patter.2025.101229.

[31] J. Burridge, et al., The inadequacy of offline large language model evaluations, Patterns 6 (12) (2025) 101397. doi:10.1016/j.patter.2025.101397.

[32] C. Ma, T. Obuchi, T. Tanaka, Neural collapse in cumulative link models for ordinal regression: An analysis with unconstrained feature model, arXiv preprint, neurIPS 2025 (camera-ready); arXiv:2506.05801 (2025).

[33] V. Y. Civale, R. Semeraro, A. D. Bagdanov, A. Magi, Intermediate layers encode optimal biological representations in single-cell foundation models, arXiv preprint, arXiv:2604.14838 (2026).

[34] K. Inecik, A. Rose, M. Haniffa, M. Luecken, F. J. Theis, Beyond visual inspection: Principled benchmarking of single-cell trajectory representations with sctram, bioRxiv preprint (2025). doi:10.1101/2025.06.23.661141.

[35] R. Martínez-Carrasco, P. Argüeso, Single-cell mapping of the glycocalyx in the adult human and mouse ocular surface, PNAS Nexus 4 (11) (2025) pgaf348. doi:10.1093/pnasnexus/pgaf348.

[36] H. He, H. Suryawanshi, P. Morozov, J. Gay-Mimbrera, E. Del Duca, H. J. Kim, N. Kameyama, Y. Estrada, E. Der, J. G. Krueger, J. Ruano, T. Tuschl, E. Guttman-Yassky, Single-cell transcriptome analysis of human skin identifies novel fibroblast subpopulation and enrichment of immune subsets in atopic dermatitis, Journal of Allergy and Clinical Immunology 145 (6) (2020) 1615–1628. doi:10.1016/j.jaci.2020.01.042.

[37] M. Farazi, S. Ramasinghe, B. S. Ciftler, M. A. Turza, S. Rahman, The gate always closes: On injecting auxiliary signals into frozen vision-language models, arXiv preprint, arXiv:2607.23335 (2026).

[38] W. Cao, V. Mirjalili, S. Raschka, Rank consistent ordinal regression for neural networks with application to age estimation, Pattern Recognition Letters 140 (2020) 325–331. doi:10.1016/j.patrec.2020.11.008.

[39] E.-I. Son, J.-T. Kim, S.-W. Seo, Ordinal neural collapse as a representation prior for visual navigation, arXiv preprint, arXiv:2606.26839 (2026).

[40] A. Aghajanyan, A. Shrivastava, A. Gupta, N. Goyal, L. Zettlemoyer, S. Gupta, Better fine-tuning by reducing representational collapse, arXiv preprint, iCLR 2021; arXiv:2008.03156 (2021).

[41] V. Papyan, X. Y. Han, D. L. Donoho, Prevalence of neural collapse during the terminal phase of deep learning training, Proceedings of the National Academy of Sciences 117 (40) (2020) 24652–24663. doi:10.1073/pnas.2015509117.

[42] G. Andriopoulos, Z. Dong, L. Guo, Z. Zhao, K. Ross, The prevalence of neural collapse in neural multivariate regression, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Vol. 37, 2024, pp. 126417–126451.

[43] A. Kumar, A. Raghunathan, R. Jones, T. Ma, P. Liang, Fine-tuning can distort pretrained features and underperform out-of-distribution, arXiv preprint arXiv:2202.10054, published at ICLR 2022 (2022).

[44] A. Merchant, D. Radev, M. Gastaldi, What happens to BERT embeddings during fine-tuning?, in: Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, Association for Computational Linguistics, Online, 2020, pp. 33–44. doi:10.18653/v1/2020.blackboxnlp-1.3.

[45] Y. Lee, A. S. Chen, F. Tajwar, A. Kumar, H. Yao, P. Liang, C. Finn, Surgical fine-tuning improves adaptation to distribution shifts, arXiv preprint, iCLR 2023; arXiv:2210.11466 (2023).

[46] E. Pareja-Lorente, P. Aloy, Task-adapted biological foundation models uncover perturbation-centric representations, bioRxiv preprint (2026). doi:10.64898/2026.06.30.735584.

[47] I. Kendiukhov, Multi-dimensional spectral geometry of biological knowledge in single-cell transformer representations, arXiv preprint, arXiv:2602.22247 (2026).

[48] P. Vig, R.-C. Wu, W. Lotter, Do pathology foundation models encode disease progression? a pseudotime analysis of visual representations, arXiv preprint, arXiv:2601.21334 (2026).

[49] K. Z. Kedzierska, L. Crawford, A. P. Amini, A. X. Lu, Zero-shot evaluation reveals limitations of single-cell foundation models, Genome Biology 26 (2025) 101. doi:10.1186/s13059-025-03574-x.

[50] Y. Zeng, et al., Cellfm: a large-scale foundation model pre-trained on transcriptomics of 100 million human cells, Nature Communications 16 (2025) 4679. doi:10.1038/s41467-025-59926-5.

[51] S. Cultrera di Montesano, et al., Improving atlas-scale single-cell annotation models with hierarchical cross-entropy loss, Nature Computational Science 6 (3) (2026) 243–249. doi:10.1038/s43588-025-00945-z.

[52] M. Hao, J. Gong, X. Zeng, C. Liu, Y. Guo, X. Cheng, T. Wang, J. Ma, X. Zhang, L. Song, Large-scale foundation model on single-cell transcriptomics, Nature Methods (2024). doi:10.1038/s41592-024-02305-7.

[53] C. Hu, C. Y.-C. Chen, Artificial intelligence virtual cells: From measurements to decisions across modality, scale, dynamics, and evaluation, arXiv preprint, arXiv:2510.12498 (2025). doi:10.48550/arXiv.2510.12498.

[54] C. Bunne, Y. Roohani, Y. Rosen, A. Gupta, X. Zhang, M. Roed, T. Alexandrov, M. AlQuraishi, P. Brennan, D. B. Burkhardt, et al., How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell (2024). doi:10.1016/j.cell.2024.11.015.

[55] L. Weidener, M. Brkić, M. Jovanović, E. Ulgac, A. Meduri, Vcbench: A multi-dimensional benchmark for single-cell foundation models, bioRxiv preprint (2026). doi:10.64898/2026.06.18.733146.

[56] S. Dou, Q. Wang, X. Qi, B. Zhang, H. Jiang, S. Chen, J. Li, M. Li, Molecular identity of human limbal heterogeneity involved in corneal homeostasis and privilege, The Ocular Surface 21 (2021) 206–220. doi:10.1016/j.jtos.2021.04.010.

[57] P. Gautam, K. Hamashima, Y. Chen, Y. Zeng, B. Makovoz, B. H. Parikh, Y. Lee, K. A. Lau, B. Lim, B. X. Ng, et al., Multi-species single-cell transcriptomic analysis of ocular compartment regulons, Nature Communications 12 (1) (2021) 5675. doi:10.1038/s41467-021-25968-8.

[58] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-R. Loh, S. Raychaudhuri, Fast, sensitive and accurate integration of single-cell data with harmony, Nature Methods 16 (12) (2019) 1289–1296. doi:10.1038/s41592-019-0619-0.

[59] Y. Du, F. Fan, S. Zhao, J. Cao, Q. Lin, K. He, T. Liu, B. Qin, M. Feng, Anchoring refusal direction: Mitigating safety risks in tuning via projection constraint, arXiv preprint, arXiv:2509.06795 (2025).

[60] tadteo, Abag-rank: attribute-based augmentation for differentiable ranking (soft-spearman implementation), GitHub repository, accessed 2026 (2025).
URL https://github.com/tadteo/ABAG-Rank

[61] E. Ramzi, N. Audebert, C. Rambour, A. Araujo, X. Bitot, N. Thome, Optimization of rank losses for image retrieval, IEEE Transactions on Pattern Analysis and Machine IntelligenceArXiv:2309.08250 (2025).

[62] S. Ali, Auditing pretraining contamination in single-cell foundation model benchmarks, arXiv preprint, arXiv:2607.20572 (2026).

[63] F. A. Wolf, P. Angerer, F. J. Theis, Scanpy: large-scale single-cell gene expression data analysis, Genome Biology 19 (1) (2018) 15. doi:10.1186/s13059-017-1382-0.

[64] K. Slowikowski, Korsunsky, Ilya, harmonypy: Python implementation of harmony, GitHub repository, accessed 2026 (2020).
URL https://github.com/slowkow/harmonypy

[65] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, A. M. Rush, Huggingface's transformers: State-of-the-art natural language processing, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, 2020, pp. 38–45. doi:10.18653/v1/2020.emnlp-demos.6.

[66] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, Pytorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems 32, 2019, pp. 8026–8037.

[67] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in python, Journal of Machine Learning Research 12 (2011) 2825–2830.
URL http://jmlr.org/papers/v12/pedregosa11a.html